

Geometric and Topological Neural Networks for Financial Applications

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Abstract

This proposal studies whether geometric and topological representations can improve alpha discovery in financial markets. The project combines order-flow features, persistent homology, graph and sheaf neural networks, and singular-learning diagnostics into a staged temporal-spatial model hierarchy. We evaluate the resulting models on daily equity data, intraday trades-and-quotes data, and level-2 or event-book data using chronological train-validation-test splits, predictive metrics, trading metrics, and robustness tests. The expected contribution is a controlled framework for measuring the marginal value of temporal, graph, topological, and sheaf-based structure for out-of-sample financial prediction.

Keywords: alpha mining, sheaf neural networks, persistent homology, Hodge and sheaf Laplacians, singular learning theory

1 Motivation and Problem Statement

Modern machine-learning models can improve return prediction, but their behavior is often difficult to interpret in non-stationary financial markets. This creates two practical problems. First, feature engineering becomes an open-ended search process. Second, model performance can be fragile when market regimes, liquidity, or cross-asset interactions change. We do not aim to eliminate feature engineering, but to replace ad hoc feature expansion with structurally motivated feature construction.

We propose to organize financial prediction models along two axes. The temporal axis describes how a model receives market information through time, from pointwise observations to attention and latent state dynamics. The spatial axis describes how a model represents relations among assets, orders, events, and higher-order interactions. The central question is whether geometric and topological structure can make this temporal-spatial hierarchy more predictive, more robust, and easier to interpret.

2 Related Work

Order-flow imbalance provides a natural microstructure feature for short-horizon return prediction and cross-impact modeling [3]. This motivates the use of dynamic order-flow features together with static order-book imbalance.

Topological data analysis offers tools for detecting persistent higher-dimensional structure in data clouds [2]. In this project, persistent homology is used as an exploratory and diagnostic layer for market-state geometry.

Sheaf neural networks extend graph-based learning by attaching local vector spaces and consistency maps to nodes and edges [4]. Connection Laplacians further enrich this framework by allowing structured transport between local representations [1]. These ideas motivate the spatial ladder from linear models to graph, simplicial, and sheaf-based architectures.

Algebraic geometry and singular learning theory provide a language for understanding model degeneracy, generalization, and the geometry of parameter spaces [6]. Recent work argues that deep-learning models are singular in a productive sense, which can help explain why over-parameterized models generalize [5]. This perspective motivates model-filtration diagnostics for under-capture, over-capture, and robustness.

3 Data and Problem Formulation

3.1 Data

The empirical study is organized into three staged benchmarks. Each stage has its own data frequency, target definition, and model class, so that additional geometric structure is introduced only after a simpler benchmark is stable.

Stage 1: daily alpha Use CRSP daily data for approximately 500 liquid assets and predict next-day cross-sectional returns or return ranks. This stage establishes a minimal benchmark with clean chronological splits, basic price-volume features, and transaction-cost-aware portfolio evaluation.

Stage 2: intraday microstructure Use TAQ or other trades-and-quotes data to construct order-flow imbalance, spread, depth, imbalance, reversion, momentum, and realized-volatility features. This stage targets short-horizon weighted mid-price movement and tests whether temporal modeling and cross-impact graphs improve microstructure prediction.

Stage 3: event-book structure Use level-2 order-book or event-book data, such as Nasdaq TotalView, LOBSTER, crypto-exchange L2 feeds, or Kalshi/Polymarket order books. This stage supports event-level queue dynamics, higher-order interactions, sheaf consistency constraints, and Hodge or sheaf-Laplacian diagnostics.

3.2 Prediction Task

Given an asset universe $\mathcal{A}$, construct features $X_{i,t}$ for each asset $i \in \mathcal{A}$ at time t . The core task is cross-sectional alpha prediction under chronological train–validation–test splits. The target is stage-dependent:

1. Stage 1 predicts next-day cross-sectional return $r_{i,t+1}$ or its rank among assets.
2. Stage 2 predicts short-horizon weighted mid-price movement, either as a direction label or a rank label.
3. Stage 3 predicts event-level price movement, queue-state transition, or cross-market response after an order-book event.

The general prediction problem is

$$\hat{y}_{i,t+h} = f_{\theta}(X_{i,t}, \{X_{j,t} : j \in \mathcal{A}\}),$$

where h is the prediction horizon and f_{θ} ranges over the temporal–spatial hierarchy described below. The chronological split prevents look-ahead bias and ensures that validation and test periods are strictly later than the training period.

3.3 Preprocessing and Feature Layers

Data cleaning Detect missing values, abnormal records, inconsistent categorical labels, and market-microstructure outliers before model fitting.

Chronological split Use time-ordered splits, with validation and test periods strictly later than the training period.

Microstructure features Include dynamic order-flow imbalance and static order-book imbalance as baseline financial features [3].

Exploratory analysis Use scatterplot matrices to inspect pairwise feature relations, outliers, nonlinear dependence, and possible regime separation.

Topological features Convert rolling market states into point clouds, build filtrations, compute persistence diagrams, and vectorize them as persistence images, landscapes, Betti curves, or persistence summaries.

Cross-asset structure Construct graphs, simplicial complexes, or sheaves from correlations, lead–lag relations, order-flow cross-impact, sector membership, and event co-occurrence.

The topological pipeline is

$$\begin{aligned} \text{rolling market states} &\longrightarrow \text{point cloud} \longrightarrow \text{persistence diagram} \\ &\longrightarrow \text{vectorized topological feature.} \end{aligned}$$

For a filtration $K_{\varepsilon}(X)$, the complex K may be instantiated as a Vietoris–Rips complex, a Čech complex, an alpha complex, a correlation-graph filtration, or a thresholded-covariance complex. The objects of interest are homology groups such as $H_n(K_{\varepsilon}(X))$.

For sheaf-based models, nodes represent assets or event types, stalks represent local feature spaces, and edge maps represent learned or estimated compatibility relations between local coordinates. The model may use a prediction loss together with a sheaf-consistency penalty so that local asset representations agree when the market relation suggests they should agree.

4 Methodology

4.1 Temporal Ladder

The temporal ladder organizes increasingly expressive ways of using history before the spatial model acts on cross-sectional structure.

Level	Temporal structure	Interpretation
T_0	$x_t \mapsto y_{t+1}$	Single-period static prediction. The model uses only the current feature vector and ignores explicit temporal history.
T_1	$(x_{t-L+1}, \dots, x_t) \mapsto y_{t+1}$	Fixed-window history. The model receives the last L observations but does not impose a learned temporal aggregation structure.
T_2	$\sum_{\tau=0}^{L-1} w_{\tau} x_{t-\tau}$	Learnable temporal kernel or convolution. The model learns stable lag weights and local temporal patterns.
T_3	$\text{Attention}(X_{t-L+1:t})$	Dynamic weighting through attention. The model adapts which historical states matter for each prediction.
T_4	State-space or recurrent latent dynamics	Hidden-state evolution, such as RNNs, GRUs, structured state-space models, Transformer state-space variants, or Kalman-style latent factor models.

4.2 Spatial Ladder

The spatial ladder organizes increasingly expressive cross-sectional structures over assets, orders, events, and market states.

Level	Model	Captured information
S_0	Linear $\hat{y}_{t+1} = w^\top x_t + b$	Single-asset linear alpha. This establishes the most interpretable baseline and captures only additive linear effects.
S_1	MLP $\hat{y}_{t+1} = f_\theta(x_t)$	Single-asset nonlinear feature interactions. This tests whether nonlinear transformations improve alpha without explicit cross-asset structure.
S_2	GNN	Pairwise asset relations, such as correlation, cross-impact, lead-lag structure, or sector graphs.
S_3	Simplicial NN	Higher-order interactions among three or more assets, sectors, or event clusters that cannot be reduced to pairwise edges.
S_4	Sheaf NN	Heterogeneous local coordinate systems and compatibility constraints across assets, suitable for relations where different nodes carry different feature semantics.
S_5	Hodge / sheaf Laplacian model	Gradient, curl, and harmonic components, used to interpret circular capital flows, local inconsistency, and global market structure.

Each upgrade is tied to a specific type of cross-sectional information, so the hierarchy tests whether additional structure is empirically justified rather than merely increasing model complexity.

4.3 Model Filtration

The temporal and spatial ladders define a two-dimensional model filtration. Each pair (T_i, S_j) corresponds to a controlled model family. The project will compare adjacent levels to determine when additional structure improves prediction and when it creates unnecessary complexity.

Along the spatial axis, model complexity forms a filtration from linear or ridge regression to multilayer perceptrons, graph neural networks, simplicial neural networks, sheaf neural networks, and Hodge / sheaf Laplacian models. The Hodge or sheaf-Laplacian layer is treated primarily as a diagnostic and regularization layer: it decomposes relational signals into gradient-like, curl-like, and harmonic components and tests whether inconsistency or circular flow improves prediction or interpretability. We then measure how performance, robustness, and interpretability fluctuate as model complexity increases. This filtration also provides a practical way to diagnose under-capture, over-capture, and robustness.

5 Experimental Design

Within each benchmark stage, each model will be trained under the same chronological data split, target definition, hyperparameter budget, and transaction-cost assumption. The primary question is not whether the largest model wins, but which structural upgrade provides measurable marginal value under a fixed experimental budget.

Stage 1	Compare linear, ridge, multilayer perceptron, and correlation-graph GNN models on daily cross-sectional alpha.
Stage 2	Add intraday order-flow features, temporal models, and cross-impact graphs for short-horizon mid-price prediction.
Stage 3	Add event-book features, simplicial structure, sheaf consistency, and Hodge / sheaf-Laplacian diagnostics for event-level prediction.
Ablation	Add one structural component at a time: order-flow features, topological summaries, graph structure, simplicial structure, sheaf structure, and Laplacian regularization.
Robustness	Evaluate performance across market regimes, asset groups, liquidity levels, and rolling test windows.
Interpretability	Study which temporal levels, topological summaries, and spatial relations contribute most to prediction, robustness, and trading performance.

6 Evaluation Metrics

The evaluation will use both statistical prediction metrics and portfolio-oriented metrics.

Prediction	Pearson information coefficient, Spearman rank information coefficient, accuracy, AUC, mean squared error, and mean absolute error.
Trading	Long-short return, profit and loss, Sharpe ratio, Sortino ratio, turnover, and transaction-cost-adjusted return.
Risk	Maximum drawdown, volatility, exposure concentration, and regime-level robustness.
Complexity	Parameter count, training stability, sensitivity to feature perturbations, marginal gain from moving up the filtration, and degradation between validation and test periods.
Diagnostics	Graph centrality stability, persistence-feature stability, sheaf inconsistency, and Hodge component energy across market regimes.

7 Expected Contributions

The project is expected to produce three contributions. First, it will provide a staged benchmark for daily, intraday, and event-book alpha prediction under consistent chronological

evaluation. Second, it will quantify the marginal value of order-flow features, topological summaries, graph structure, simplicial structure, sheaf consistency, and Hodge / sheaf-Laplacian diagnostics. Third, it will connect empirical model selection with geometric and singular-learning diagnostics, focusing on where additional structure improves robustness and where it merely increases complexity.

8 Timeline and Risks

Phase 1	Build the CRSP daily data pipeline, define daily return and rank targets, and implement chronological splits.
Phase 2	Train daily baselines and graph models, then establish prediction, trading, and risk benchmarks.
Phase 3	Extend to TAQ intraday data with order-flow features, temporal models, and cross-impact graph ablations.
Phase 4	Extend to level-2 or event-book data and evaluate topological, simplicial, sheaf-based, and Hodge / sheaf-Laplacian modules.
Main risks	Data sparsity, asynchronous sampling, transaction-cost sensitivity, overfitting, and unstable regime behavior. These risks will be mitigated through rolling-window validation, strict ablation, stage-specific targets, and regime-specific evaluation.

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